

LAB EXPERIMENT RECORD**08-10-2026---Set 15: Product Sales Analysis**

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Date	08 OCTOBER 2026
Day	Thursday

Aim

To analyze the first-quarter sales of each product using a grouped bar chart, stacked area chart and table, and to build an interactive Tableau dashboard for exploring the sales data.

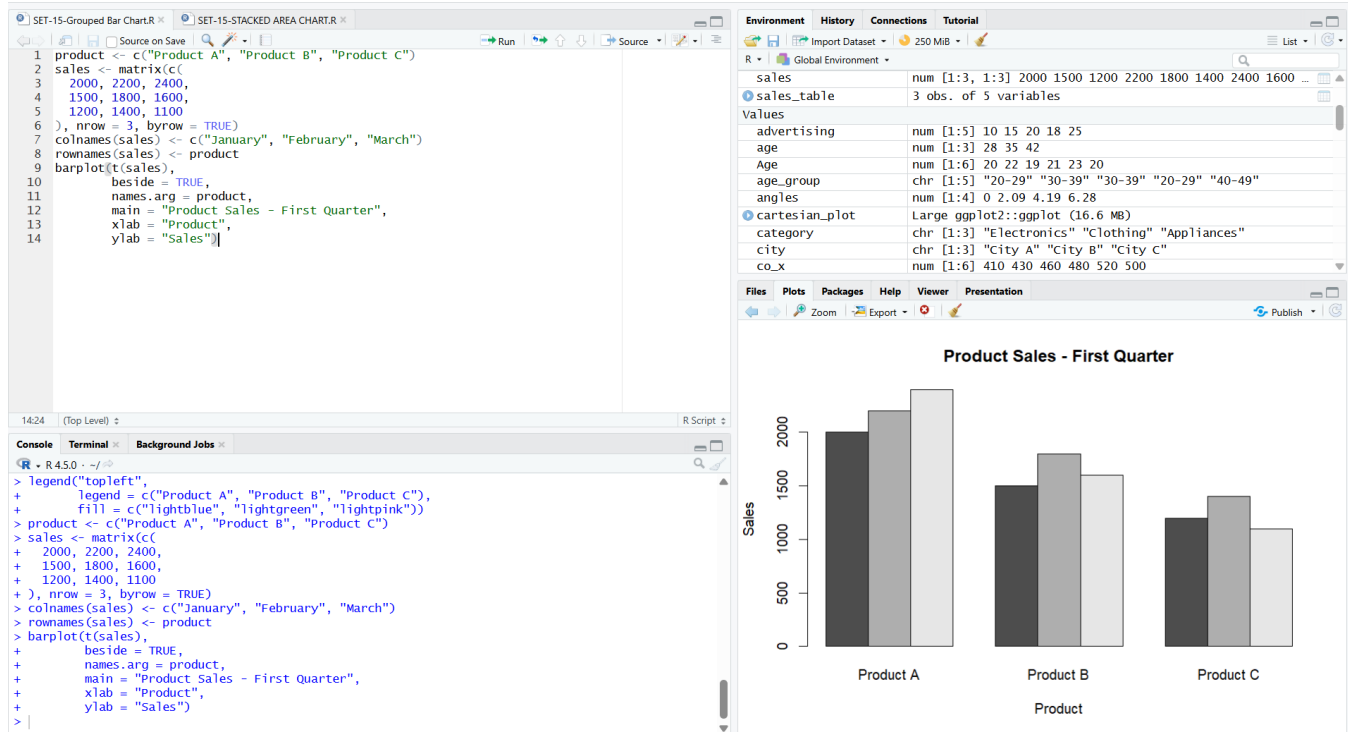
Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

Question 1: Grouped Bar Chart of First-Quarter Sales

Question: Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

Aim: To compare the January, February and March sales of each product using a labelled grouped bar chart.

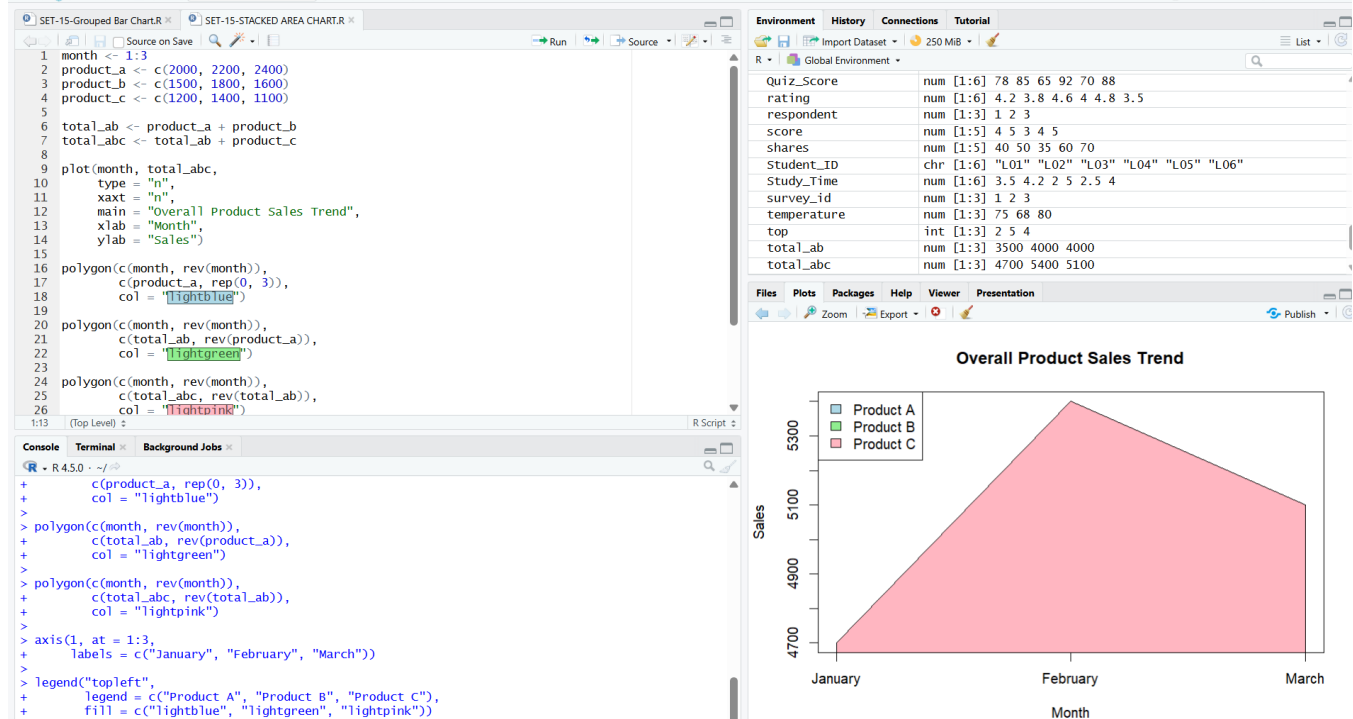


Result: Product A has the highest sales in every month, rising from 2000 to 2400, while Products B and C peak in February.

Question 2: Stacked Area Chart of Overall Sales Trend

Question: Generate a stacked area chart to represent the overall sales trend for all products over the three months.

Aim: To represent the overall sales trend of all products over the three months using a stacked area chart.



Result: Combined sales rise from 4700 in January to a peak of 5400 in February, then fall to 5100 in March.

Question 3: Table of Monthly Product Sales

Question: Build a table to show the monthly sales data for each product. Label the table elements.

Aim: To present the monthly sales of each product in a labelled table.

The screenshot shows the R Studio environment. The top pane displays the following R code:

```

1 #BANDLAPALLI BHANUTEJA REDDY
2 product_id <- c(1, 2, 3)
3 product_name <- c("Product A", "Product B", "Product C")
4 january <- c(2000, 1500, 1200)
5 february <- c(2200, 1800, 1400)
6 march <- c(2400, 1600, 1100)
7
8 sales_data <- data.frame(
9   Product_ID = product_id,
10  Product_Name = product_name,
11  January_Sales = january,
12  February_Sales = february,
13  March_Sales = march
14 )
15
16 print(sales_data)

```

The bottom pane shows the console output of the code:

```

> january <- c(2000, 1500, 1200)
> february <- c(2200, 1800, 1400)
> march <- c(2400, 1600, 1100)
>
> sales_data <- data.frame(
+   Product_ID = product_id,
+   Product_Name = product_name,
+   January_Sales = january,
+   February_Sales = february,
+   March_Sales = march
+ )
>
> print(sales_data)
  Product_ID Product_Name January_Sales February_Sales March_Sales
1          1   Product A         2000          2200         2400
2          2   Product B         1500          1800         1600
3          3   Product C         1200          1400         1100

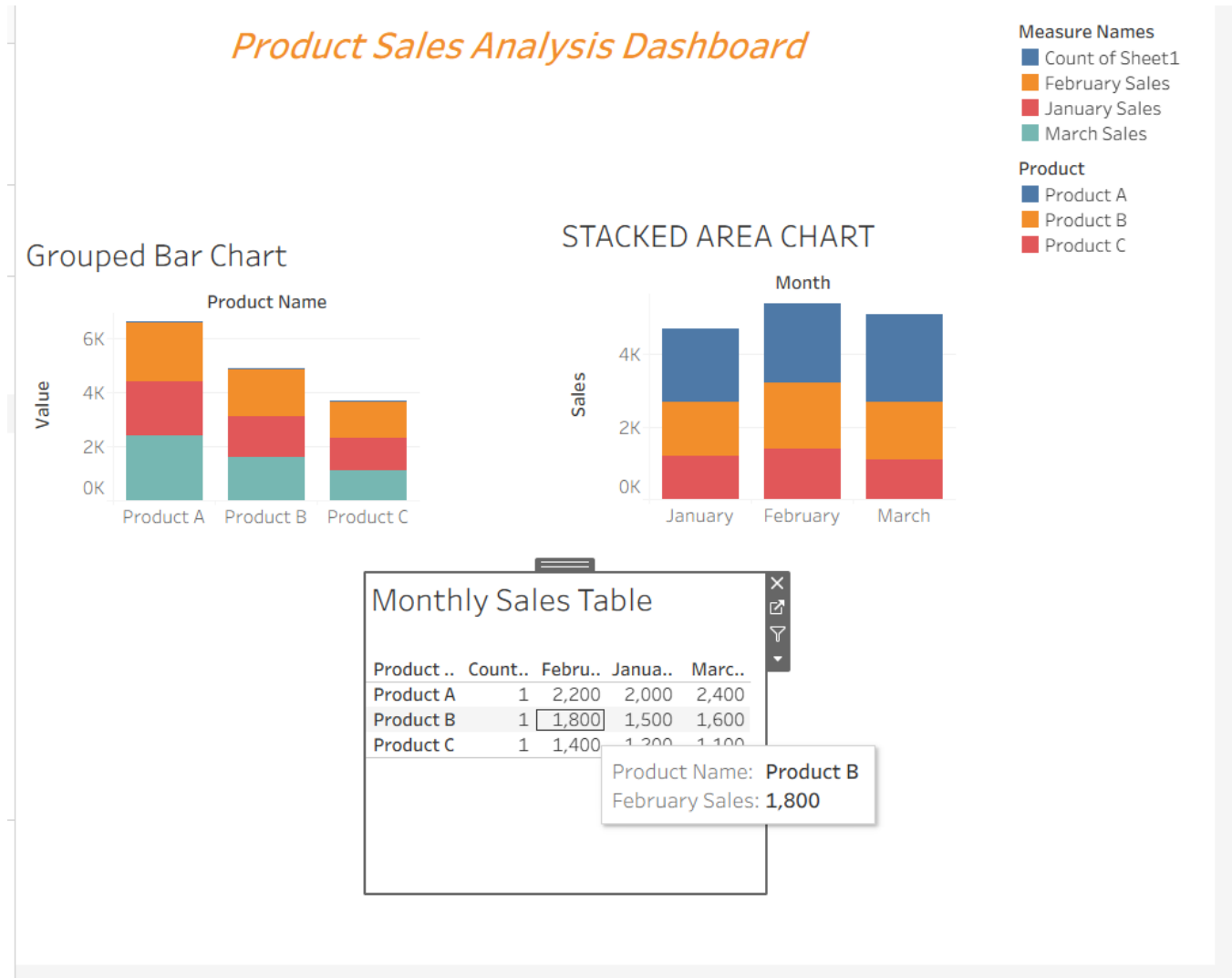
```

Result: The table lists each product's monthly sales; Product A totals 6600, Product B 4900 and Product C 3700 for the quarter.

Question 4: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

Aim: To build an interactive Tableau dashboard that combines the grouped bar chart, stacked area chart and table for exploring sales data.



Result: The dashboard places all three views together, and filters on product and month let users explore sales interactively.

Overall Result

Observation: Total first-quarter sales were 15200, of which Product A contributed the most (6600), and February was the best month overall (5400).